

**ROUGH GAPS EXIST? OPENING GAPS HELPS TO SURGE
RETURNS IN SWING AND INTRADAY TRADING.**

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Abstract

This study examines the importance of fundamental price anomalies in the American stock market, specifically examining the S&P 500 and NASDAQ indexes spanning. The study seeks to provide an understanding of gap trading behaviour and consequences for trading strategies, particularly profit reduces risks in optimization and swing and intraday trading. The study uses a comprehensive approach, including a pre-test methodology and post-test analysis, to examine gap size, trend, and closure from 2019 to 2021, using a data set of 750 samples.

The study also examines three hypotheses: first, that larger gap increases the likelihood of market inversions; Second, there is a correlation between the large gap and the trading volume and third, that gap trading behaviour may affected by economic variables like gold and the US dollar. According to the findings, there is a significant correlation between gap size and market reversals because big gaps tend to result in bigger market retractions. There is only a poor correlation between gap size and trading scale, so other independent variables may influence trade flow. Most importantly, there is only weak relationship between gap behaviour and economic variables like gold prices or dollar value.

It emphasizes the nature of effective market thinking and comprehensive business planning that goes beyond just basic economic issues. This research provides us beautiful pictures of gap analysis and a new way to look at the gaps, which predicts the direction in which market will move. Secondly it is relevant on how we would take note upon other components such as social aspect, psychological aspects among others while deciding concerning trades. To enhance our comprehension of market dynamics, future research should seek those analysing global markets and machine learning models - many economic indicators included.

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Chapter 1

1. Introduction

Gap analysis is an important method in stock market research that focuses on understanding and analysing stock price behaviour after gap openings. A gap is a discontinuity in the price chart of an asset where the price leaps higher or lower from the previous day's closing with no activity in between (CHEN, 2021). Common gaps, breakaway gaps, runaway gaps, and fatigue gaps are the four types of gaps (CHEN, 2021). These gaps develop as a result of news or events that cause a rush of buyers or sellers, perhaps suggesting the beginning of a new trend or the reversal of a prior trend (CHEN, 2021).

The purpose of this research article is to investigate the price gap anomaly in the US stock market, especially the S&P 500, and NASDAQ indexes from 1928 to 2018 (Aiche Avishay, Cohen and Griskin Vladimir, 2023). We want to obtain insights into the consequences of these gaps for traders and investors by analysing the price behaviour of equities after gap openings in both upward and downward directions. Understanding the features and repercussions of various types of gaps can help market players establish effective profit and risk management strategies. (Aiche Avishay, Cohen and Griskin Vladimir, 2023)

We hope to contribute to the current body of knowledge in technical analysis through this research by offering a detailed assessment of gap analysis and its relevance in anticipating market developments (Shankar, Miller and Balakrishnan, 2020). Traders and investors can enhance their overall performance in the financial markets by understanding the ideas and procedures of this strategy (Shankar, Miller and Balakrishnan, 2020).

1.1. Research Problem

As the research problem, this study seeks to provide a deeper understanding and strategic approach for trading opening price gaps oriented towards returns maximization measures while optimally managing risks in the dynamic market environment.

1.2. Aims and Objectives.

Aim

The dissertation seeks to provide answers to the questions of traders that they should trade gaps to increase profits while accumulating wealth through intraday trading and investment. With the help of this research, we can draw the conclusion that there are various types of trading strategies that can assist both institutional and retail traders in making money by trading opening gaps based on the behaviour of identifying gaps. The primary motivation for this gap analysis is that the majority of traders overlook the fact that gaps may be traded and that they exhibit patterns that enable traders to earn higher profits.

Objectives

1. Determine which gap can be traded, as well as when to enter and exit the gap;
2. To analyse price gaps in relation to their magnitude, direction, trade volume, and their impact on trading success
3. Analyse the performance of the S&P 500 during the past three years (2019–2021) to carefully assess the performance of the opening gaps.
4. Create suggestions for investors on how to use gap trading to increase returns.

1.3. Hypothesis for test

H1: Impact Of Gap Magnitude on Market Reversal

Null Hypothesis (Ho): The magnitude of the opening gap (as a percentage change) does not significantly affect the likelihood of a market reversal from the gap direction.

Alternate Hypothesis (H1): The magnitude of the opening gap (as a percentage change) significantly affects the likelihood of a market reversal from the gap direction.

H2: Correlation between Gap Size and Trading Volume

Null Hypothesis (Ho): There is no significant correlation between the size of the opening gap and the trading volume on the day the gap occurs.

Alternate Hypothesis (H1): There is a significant correlation between the size of the opening gap and the trading volume on the day the gap occurs.

H3: Influence of Economics Indicators (Gold & USD)

Null Hypothesis (Ho): Economic indicators such as gold prices and USD value do not significantly influence the behaviour of opening gaps.

Alternate Hypothesis (H1): Economic indicators such as gold prices and USD value do not significantly influence the behaviour of opening gaps.

Chapter 2

2. Literature Review

2.1. Introduction

In this chapter, we present a comprehensive literature review on the intersection between psychological factors and global economic events in terms of gap trading. The trading world does not consist of only tangible indicators and occurrences that have to do with the economic but is also defined by the mentality of all stakeholders in the market. Understanding these conceptual principles is important to better appreciate the complexity of trading strategies, especially under the globalized and interdependent nature of today's financial markets.

We begin by studying the psychological aspects of gap trading, focusing on the influence of bias, emotion, and cognitive mechanisms on trader behaviour and market outcomes. This study draws on renowned researchers such as Kuchta (2023), Nathanson (2019), and Blandin & Anderson (2023), who have delved into the consequences of traits such as overconfidence, stereotyping and availability bias in the workplace.

After that we dive into how macroeconomic activities such as supply chains and changes to key financial signs affect the conduct on markets where traders take advantage. This area relies on the results from studies that are conducted by the European Central Bank and other knowledgeable people who have analysed the huge impact of world affairs in financial markets.

In the field of fast-changing movements and looking into gap trading in future we explore social media's role in shaping investor behaviour patterns. However, we are searching for the role platforms such as Twitter and Reddit have acquired in a broader sense also tend to be formidable catalysts of market sentiment leading to quick oscillations on the markets.

The last part of the chapter discusses how social media data have been introduced into algorithmic trading, which is transforming the very nature of modern financial market. This section focuses on the pros and cons of using social media sentiment in trading algorithms, highlighting that supervision is very necessary to minimize the false information propagation as well as manipulation within a market. After reading this chapter, the readers will have a good understanding of the complex interaction between psychological factors, world economic incidents and upcoming trends in the gap trading. With this knowledge, we will have a very solid foundation to proceed with many practical trading strategies and understanding the market forces that dictate various investment decisions in today's digital interconnected world.

2.2. Strategies for Putting and Exiting Price Gaps in Trading

Both traders and analysts have long been enthralled by the occurrence of price gaps in financial markets. This literature study explores the topic of trading gaps, concentrating on the gaps that can be traded successfully as well as the best times to initiate and exit such transactions. The main objective is to identify methods that enable traders to increase earnings and build wealth through intraday trading and investing, illuminating frequently disregarded patterns related to gap behaviour. When there is a noticeable difference between the closing price of one trading session and the opening price of the next, price gaps become apparent. Various market variables, such as news releases, economic data, and overnight events, might result in such gaps. Although there is unquestionable potential for profit from gap trading, careful analysis is necessary to successfully traverse the challenges of this approach (Zarattini and Aziz, 2023).

Extensive research has been conducted to determine which gaps are most conducive to effective trading in the quest for useful insights into gap trading. Based on the characteristics of the gap, one classification has been developed. Traders can better grasp the context and effects of certain gaps by using a typology that includes

Common Gaps, Breakaway Gaps, Continuation Gaps, and Exhaustion Gaps. Breakaway Gaps, for instance, signal the start of a new trend and can present profitable opportunities for traders who take advantage of the momentum. Contrarian traders may choose to explore positions against the current trend if Exhaustion Gaps, on the other hand, indicate the peak of a trend.

The gap trading strategic decision-making process is greatly influenced by the size and direction characteristics. Larger gaps, those brought on by noteworthy news or events, have the ability to bring about more noticeable price changes. Such gaps serve as powerful market sentiment indicators, frequently sparking trends that may last past the current trading day. Both bullish and bearish gaps are crucial, as their directional implications provide information about whether current trends will likely continue or reverse. Trading professionals can gain vital insights that enable them to make well-informed entry and exit decisions by carefully studying these criteria. Through a better grasp of market dynamics, this strategy not only increases the possibility of winning trades but also helps with risk management. In gap trading, trade volume serves as a validating process, providing crucial confirmation of a gap's likelihood of success. Increased volume during gap periods emphasizes the likelihood of price movement in the gap's direction and provides weight to the gap's validity (Walker, 2021).

The volume-gap success correlation can help traders feel much more confident about their trading choices. With this confirmation, traders are better equipped to fine-tune their methods, maximizing risk-reward ratios and profiting from market trends. By incorporating volume analysis into gap trading tactics, traders are better able to manage the complexities of market behaviour, turning what first appear to be random price gaps into profitable trading opportunities. Gaps play an important role in price movement, but they also operate as strong indicators of levels of support and resistance. After it forms, a gap that is left unfilled and untested might develop into a strong level of support or resistance, which will affect how the market behaves moving

forward. By making it easier to set stop-loss orders and profit objectives, this understanding gives traders a tactical advantage. Profiting from possible advantages while utilizing this knowledge, traders can safeguard their investments from unfavourable market changes. By strategically using gaps as support and resistance levels, traders are better able to manage their overall risk and make decisions that are in line with their trading goals.

The performance of the S&P 500 over the last three years (2019–2021) is the focus of the research given the aforementioned factors. This information can be used to gain important understanding of opening gap behavior and how it affects market movements. In the context of the S&P 500, categorizing gaps by type, direction, and amplitude makes it easier to evaluate the efficacy of gap trading techniques thoroughly. Investors can get specialized advice to boost returns if they have a good understanding of gap dynamics and their impact on trading success. Gap analysis must be combined with complimentary technical and fundamental analyses to create a successful gap trading strategy. This all-encompassing strategy gives traders a comprehensive toolkit they may use to make well-informed decisions and take advantage of market chances. Gap trading is a practical approach that offers untapped potential as traders navigate unpredictable financial markets. Although gaps bring opportunities, it is crucial to recognize that they also carry risks. Due to the volatility in the vicinity of gap areas and the potential for a gap retracement, careful risk management is necessary using tools like stop-loss orders (Guo et.al., 2020). The literature's examination of gap trading highlights how successful an undertaking it may be. Traders can more confidently negotiate the complexities of gap trading by selecting the sorts of gaps to trade, assessing their magnitude, direction, and trade volume, and understanding their impact on trading performance. These insights are strengthened even further by the empirical analysis of S&P 500 performance over a specified time period, which directs investors toward strategies that fit their objectives and risk

tolerance. The strategic use of gap trading may become an important strategy for traders looking to increase profitability and build riches as the financial markets continue to change.

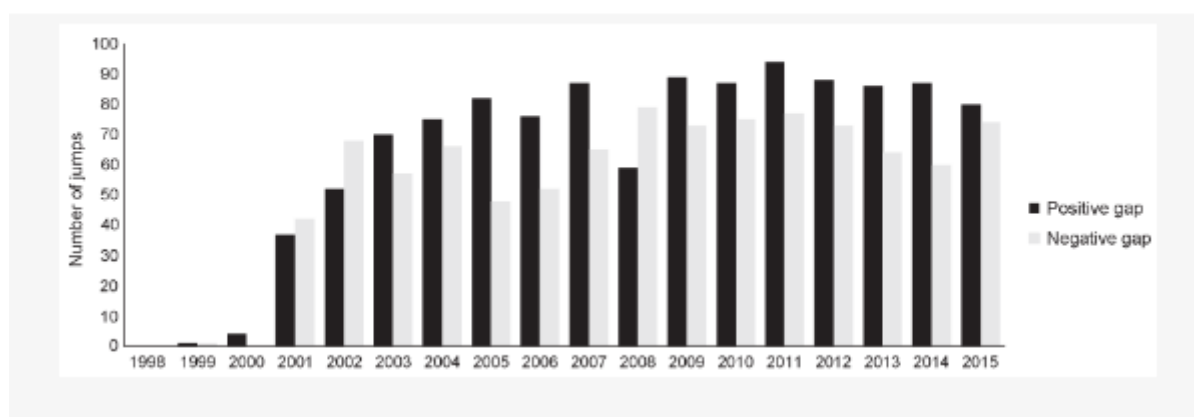


Figure 1 Development of positive and negative overnight gaps as identified by the BNS jump test, from 1998–2015.

Chan and Fong (2019) provide a statistical arbitrage technique based on the hypothesis of mean-reverting jump-diffusion and apply it to S&P 500 high-frequency component data from January 1998 to December 2015. The specific technique spotted overnight price gaps using an advanced jump test process and took advantage of market abnormalities that occur during the initial minutes of a trading day. The approach generates statistically and economically meaningful returns of 51.47% p.a. and an annualised after-transaction cost Sharpe ratio of 2.38. The paper's graphic above depicts the number of recorded nocturnal gaps over time. Rising volatility in financial markets causes a rise in the frequency of overnight gaps; market variances simply surge. As predicted, no jumps were seen in the initial years of this study's sample. In contrast, the number of overnight price gaps grew during the period of significant market volatility. Positive gaps often have a greater influence on the S&P 500 index than negative gaps. The graph shows how the pattern shifts during crises such as the early 2000s dot-com disaster and the 2008 financial crisis.

2.3. Analysing Price Gaps: Direction, Magnitude, Volume, and Impact on Trading Success

A key component of trading techniques and decision-making is the examination of price gaps on financial markets. The goal of this study is to explore the complex link between price gaps, their characteristics (magnitude, direction, trade volume), and how these factors affect trading success. This review clarifies the complex dynamics that drive gap trading and its potential consequences for traders looking to maximize profitability by examining these aspects in the context of existing literature.

2.3.1. Magnitude of Price Gaps

The study conducted by Johnston, and Land-Kazlauskas (2018), highlights the size of a price gap becomes a dynamic force in the trading world, reshaping the terrain that traders must negotiate. Although the idea of magnitude—the numerical difference between closing and opening prices—seems simple, it has significant consequences that go far beyond the number. The field's literature emphasizes the critical importance of wider gaps, which frequently appear in response to significant news releases or unexpected events. A more in-depth story about market behaviour is revealed within these bigger gaps. Larger gaps have a unique tendency to draw market participants' attention because of their large magnitude. They act as visible indicators of increased market volatility, indicating a change from the typical ups and downs of prices (Tushar et.al., 2019). While potentially frightening, this volatility offers traders a special chance to capitalize on market momentum. The research sheds light on the notion that these greater gaps are not just outliers but rather powerful catalysts capable of inducing substantial changes in price dynamics. Researchers frequently discover the existence of market-moving information—earnings reports, macroeconomic data, geopolitical developments—that causes prices to soar into uncharted territory when analysing the causes of these gaps. One of the noticeable characteristics of these large gaps is their propensity to start trends that last much longer than the current trading session. Researchers have been attracted by this phenomenon because they believe that

these gaps herald the beginning of trends that could influence market trajectories over a longer time frame (Johnston, and Land-Kazlauskas, 2018).

This realization adds a strategic component to gap trading, one that goes beyond the quick transactions' instant delight. Where author El Ghorayeb (2021) said that the Traders who are able to accurately spot and align their positions with these momentum-driven trends stand to benefit from increased returns that result from the trends' durability. The temptation of trading bigger gaps is not without risk, though. The same volatility that drives prices to record highs also makes it possible for abrupt and quick reversals. These gaps are vulnerable to unexpected changes in sentiment due to the volatile nature of the financial markets, which could potentially skew expected price moves. This inherent risk necessitates a well-structured risk management strategy—one that respects the necessity for safety nets against unanticipated reversals while acknowledging the attractiveness of wider gaps. It is crucial for traders to understand that while these gaps offer tempting opportunities, they must also be treated with a careful awareness of the associated hazards. The size of a price gap encompasses a complex interplay of volatility, momentum, and the possibility of trend start; it is far from being a static numerical figure (Zarattini and Aziz, 2023). Larger gaps are indicative of dynamic movements in market sentiment and are frequently triggered by news and events with high effect. Traders have an opportunity for increased rewards due to their capacity to launch long-lasting trends. However, this potential for profit is accompanied with increased volatility and the risk of unanticipated reversals. A sophisticated technique that incorporates these characteristics with a well-thought-out risk management strategy becomes essential as traders consider the appeal of trading greater gaps (Siegel, 2021).

2.3.2. Direction of Price Gaps:

According to the author Pedersen (2019), the direction of price gaps appears as a guiding light, pointing traders toward potential trends and movements, inside the

complex terrain of financial markets. The two poles of gap direction, bullish and bearish gaps, have the ability to shed light on the direction of market trends. A bearish gap suggests the probability of a trend reversal, whilst a bullish gap indicates the continuation of a continuous upward trend. Both researchers and practitioners have noticed the importance of this directional information. The predictive power of the gap direction is highlighted by extensive study. These directional cues offer important information about prospective price changes and act as early indicators of market sentiment. Gap direction improves the accuracy of trading decisions when it is weaved into the fabric of technical and fundamental analysis. Traders that have a keen awareness of gap direction are better able to negotiate the complexity of market dynamics, matching their positions with developing trends for the best possible results (Gereffi, 2019). It's important to understand that although gap direction has predictive value, it works best when combined with other analytical methods. Using the gap direction alone may result in insufficient insights. The likelihood of making correct predictions is increased by the interaction between gap direction and technical indicators, chart patterns, and fundamental variables. Gap analysis and a wide range of tools are combined in this synergy, enabling traders to make more thoughtful and educated selections. Gap direction plays a dual role in the complex patchwork of trading methods, acting as both a core component and a supplementary one. Because of its function as a compass for navigating market trends and the recognition of its symbiotic relationship with other analytical tools, it gives traders the ability to understand market nuances and capture opportunities as they arise. To realize gap direction's full potential in the dynamic world of financial markets, a holistic analytical approach must be combined with it (Nti, Adekoya, and Weyori, 2020).

S&P 500 Index (\$SPX)

4,405.71 +29.40 (+0.67%) 08/25/23 [INDEX/CBOE]

PERFORMANCE REPORT for Fri, Aug 25th, 2023

Alerts  Watch  Help 

Past 5 Days							Daily
Date	Open	High	Low	Last	Change	%Chg	Volume
08/25/23	4,389.38	4,418.46	4,356.29	4,405.71	+29.40	+0.67%	0
08/24/23	4,455.16	4,458.30	4,375.55	4,376.31	-59.70	-1.35%	0
08/23/23	4,396.44	4,443.18	4,396.44	4,436.01	+48.46	+1.10%	0
08/22/23	4,415.33	4,418.59	4,382.77	4,387.55	-12.22	-0.28%	0
08/21/23	4,380.28	4,407.55	4,360.30	4,399.77	+30.06	+0.69%	0

Figure 2: Performance Report

(Source: Barchart, 2023)

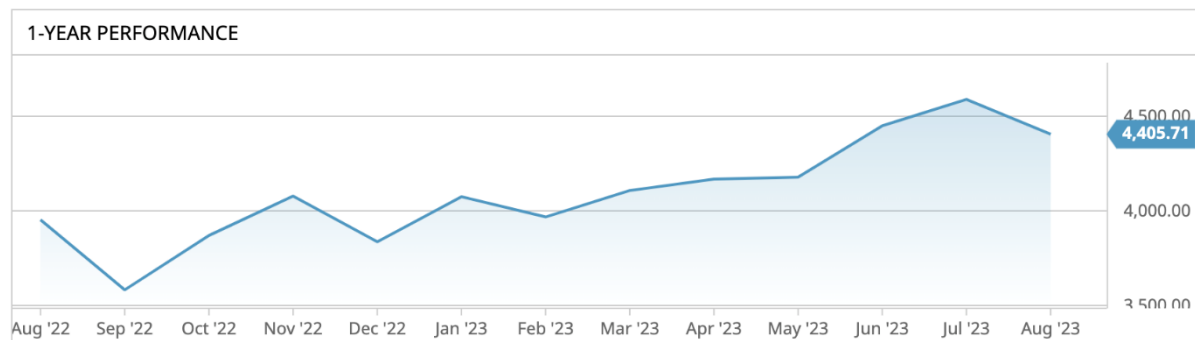


Figure 3: 1 Year Performance

(Source: Barchart, 2023)

2.3.3. Trade Volume in Gap Analysis:

Similarly, study conducted by Feng et al., (2023), highlights trade volume emerges as a compelling narrative that strengthens the justification and success of gap trades in the complex tapestry of gap research. It acts as a crucial corroboration factor that confirms a gap's veracity and highlights its potential impact on price dynamics. Researchers and traders alike have been fascinated by this interaction between trade volume and gaps because it sheds light on the motivations and actions of market players. The increase in trading volume that occurs when gaps arise reflects the

eagerness with which market players react to new price moves. A quick change in market mood caused by the influx of fresh information or events frequently fuels this increased trading activity. The assumption that increased trading volume is a measurable manifestation of market participants' confidence in the gap's direction permeates the literature. This vote of confidence strengthens the likelihood that the ensuing price movement will endure. Trading techniques can be improved by taking advantage of the symbiotic link between trade volume and gap authenticity (Feng et al., 2023).

While authors In, Rook, and Monk (2019), define that traders can better comprehend the dynamics underpinning gap creation by incorporating volume analysis into their decision-making. Trading professionals can more precisely tune their entry and exit locations by using volume as a confirming indication. By doing this, individuals can successfully reduce the danger of entering trades without the required market support, increasing their chance of trading success. It's important to understand that although transaction volume is a potent confirming indication, its interpretation calls for deftness. An intricate method of volume analysis takes into account both the volume in later sessions as well as the volume during the gap itself (Zarattini and Aziz, 2023). A more thorough picture of the shifting market sentiment and the persistence of the price movement caused by the gap is provided by this wider viewpoint. The combination of gap analysis and trade volume insights adds a layer of depth and confidence to the always changing world of trading techniques. This integration supports traders' overarching goal of making well-informed decisions that optimize profitability while lowering risk. Understanding how trade volume and gap authenticity interact allows traders to better capitalize on opportunities that are in line with the mood of the market, boosting their chances of trading success in a field that is becoming more and more competitive (In, Rook, and Monk, 2019).

2.3.4. Impact on Trading Success

According to the authors Strawser, Hechavarría, and Passerini (2021), price gaps have an impact on trading performance that goes beyond their immediate price impact by weaving a complex story that influences upcoming market behaviour. Gaps have the natural ability to transform into crucial benchmarks that control the ebb and flow of price patterns, known as support and resistance levels. Researchers and traders are drawn to this latent potential because it signals the strategic opportunities that result from gaps' capacity to shape market trajectories. Particularly unfilled and untested gaps transform into powerful support or resistance zones that act as anchors for later market movements. These zones serve as benchmarks against which price changes are evaluated, giving traders a vantage point from which to determine critical aspects. The traders' risk management tactics are strengthened by carefully lining up stop-loss orders and profit targets with these zones. The development of the gaps from simple price voids to dependable support and resistance levels adds a degree of assurance, enabling traders to deal with market volatility using a well-organized strategy (Vos, Martin, and Resnick, 2023).

The key to successful trading is this dynamic synergy between gap transformation and efficient risk management. A thorough trading strategy that incorporates gaps as reference points gives the procedure a strategic depth. Researchers stress that a number of things must come together for trading success, including a keen eye for spotting and analysing gaps, a methodical approach to risk management, and a wider strategy that takes into account risk tolerance, portfolio diversification, and current market conditions. Gaps become more than just anomalies in this all-encompassing approach to trading; they develop into strategic anchors that tie traders to good outcomes. Trading professionals can optimize their risk-reward ratio and improve the accuracy of their decision-making by taking advantage of gaps' ability to develop into support and resistance zones. A thorough trading approach that takes advantage of

gaps enhances profits while protecting capital. The incorporation of gaps into this overarching approach, as the trading environment changes, is evidence of their continuing importance in achieving trading success (Rout, 2018).

Figure 4 Estimates of shadow cost before and after index changes

Table V: Estimates of Merton's Shadow Cost

Merton's shadow cost is defined as below:

$$\text{Shadow Cost} = \frac{\text{Residual Standard Dev}}{\text{S \& P 500 Market Cap}} \times \frac{\text{Firm Size}}{\text{Number of Shareholders}}$$

where Firm Size (the market value of the firm's equity) and S&P 500 Market Cap are measured on the announcement day. The residual standard deviation is measured as the standard deviation of the difference between the firm's return and the total return of the S&P 500 index in the 252-days period before the index change announcement day (pre), and in the 252-day period after the effective day (post). The number of shareholders before the index change is obtained as late as possible prior to the announcement date. The number of shareholders after the change is obtained at least nine months after the effective date.

Period	Addition / Deletion	Sample Size in Table I	Sample Size for this Table	Mean (Median) shadow cost Before Change (x 10 ⁹)	Mean (Median) shadow cost After Change (x 10 ⁹)	Mean (Median) of Paired Changes
197609-198909	Additions	263	243	2.175 1.491	1.695 1.337	-0.480*** -0.095***
	Deletions	28	20	0.283 0.227	0.313 0.247	0.030 0.010
198910-200012	Additions	218	187	11.537 3.595	7.956 3.040	-3.581*** -0.054***
	Deletions	62	54	0.450 0.256	0.563 0.278	0.112* 0.055***

*: Significant at 0.10 level

**: Significant at 0.05 level

***: Significant at 0.01 level

Harris and Gurel (2003) explore the price reaction to S&P 500 index additions and removals in their article. The findings support the hypothesis that changes in investor knowledge contribute to the asymmetric price impact of S&P 500 index additions and removals. Figure 4 depicts the estimates of shadow cost before and after index modifications. The mean shadow cost decreased by 0.48 from 1976 to 1989, and additions decreased by 3.58 from 1989 to 2000. For deletions, the shadow cost increased by 0.11 between 1989 and 2000.

2.4. Analysing the Performance of the S&P 500 Opening Gaps (2019-2021)

By the study of author Ramodike (2022), Investors and analysts alike have paid close attention to the S&P 500 index's performance over the past three years (2019–2021). The benchmark S&P 500 index, which includes 500 of the most significant US stocks, provides helpful information on the state and developments of the US stock market. This time period, which is characterized by both economic stability and volatility, offers a favourable environment for researching opening gap behaviour and how it affects the performance of the index. The three-year return of the S&P 500 is used as a quantitative indicator of the index's performance over this time. This return, at 40.29%, is higher than both the long-term average of 23.09% and the one-year return, which is 38.58%, demonstrating a healthy and upward-trending market (Ycharts.com, 2023). However, one needs go further into the connection between these gaps and the index's overall performance in order to fully understand the significance of opening gaps. Interesting patterns can be seen by looking at the data. While authors Balcilar, M., Ozdemir, Z.A. and Ozdemir (2021) highlights that it is noteworthy that the S&P 500 has demonstrated various levels of volatility, as seen by changes in its monthly returns. These variations serve as the basis for gap analysis since they are impacted by a wide range of variables, such as economic indicators, geopolitical developments, and business earnings reports. Opening gaps, which represent differences between the closure of the previous day and the open of the current day, stand out as interesting statistics in this context. Their occurrence may be a result of recent events, changes in the world, or changes in investor attitude. The data shows that the S&P 500 is sensitive to these gaps, with returns changing dramatically as a result. The S&P 500's performance in respect to opening gaps over this time has been complex. Larger gap months frequently coincide with big price changes.

Positive returns are usually correlated with bullish gaps, indicating long-term rising trends. Similar to bullish gaps, bearish gaps might indicate market declines. However,

there isn't necessarily a linear correlation between gaps and eventual returns. Although wider gaps often indicate higher price fluctuations, other elements including volume, economic indicators, and outside events also come into play (Walker, 2021). The data inspires more research into the subtleties of these interactions. For traders looking to take advantage of these patterns, correlations between gap size, direction, and future returns may provide useful information. For instance, trading in the direction of wider gaps can result in more profits. However, author Doran believe that due to the market's complex dynamics, care is advised since they may cause unexpected reversals and departures from historical trends. Opening gaps must be included in a more comprehensive trading plan in order to be effective as prediction tools. Even if they are instructive, these gaps shouldn't be seen in isolation. Technical and fundamental research, market sentiment, and risk management are all included in a holistic strategy. Opening gaps at this point add to a trader's toolset by improving decision-making accuracy and providing prospective opportunities to profit from developing trends. Analysis of the effect of opening gaps on trading success may be done against a compelling backdrop provided by the performance of the S&P 500 over the last three years. The index's overall rise is highlighted by the strong three-year return, but it is the interaction between these returns and the incidence of opening gaps that reveals an engrossing story. Although wider gaps frequently coincide with significant price changes, their successful incorporation into trading methods requires a comprehensive strategy that takes into account a number of different elements. Opening gaps act as instructive data points that indicate traders in the direction of prospective opportunities, but their potential must be wisely tapped within a thorough trading framework. Examining opening gaps is a crucial component of strategy building for traders trying to negotiate the constantly shifting landscape of the financial markets as market dynamics continue to change (Collins, and Halverson, 2018).

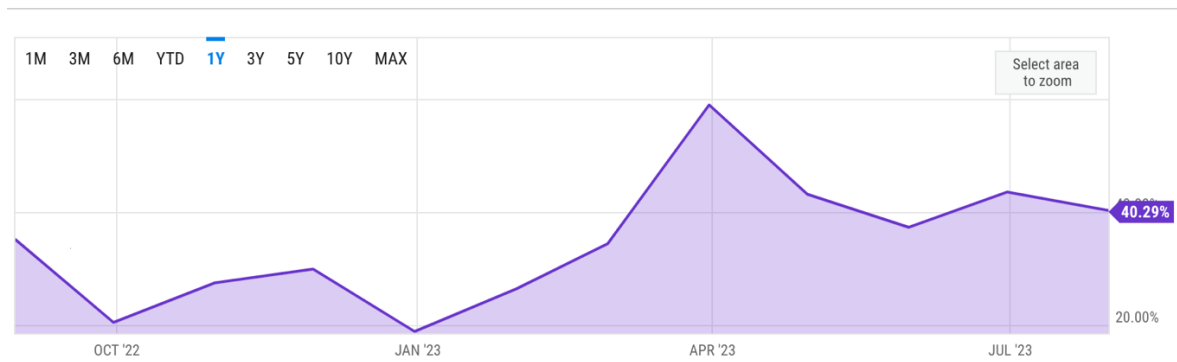


Figure 5: S&P 500 3 Year Return Chart

(Source: Ycharts, 2023)

Stats

Last Value	40.29%
Latest Period	Jul 2023
Last Updated	Aug 5 2023, 11:02 EDT
Next Release	Sep 5 2023, 11:00 EDT (E)
Long Term Average	23.09%
Average Growth Rate	-82.77%
Value from Last Month	43.55%
Change from Last Month	-7.49%
Value from 1 Year Ago	38.58%
Change from 1 Year Ago	4.42%
Frequency	Monthly
Unit	Percent
Adjustment	N/A
Download Source File	Upgrade

Figure 6: according to states

Related Indicators	
S&P 500 Fundamentals	
S&P 500 Dividend Yield	1.54%
S&P 500 Earnings Yield	4.26%
S&P 500 Market Cap	37.16T USD
S&P 500 P/E Ratio	23.46
S&P 500 Price to Book Ratio	3.748
S&P 500 Shiller CAPE Ratio	30.82
SPX Put/Call Ratio	1.39
VIX	15.68
S&P 500 Returns	
S&P 500 1 Year Return	11.11%
S&P 500 10 Year Return	172.2%
S&P 500 5 Year Return	62.94%
S&P 500 Annual Return	-19.44%
S&P 500 Annual Total Return	-18.11%
S&P 500 Monthly Return	3.11%
S&P 500 Monthly Total Return	3.21%

Figure 7: Related Indicators And S&P 500 Returns

(Source: Ycharts, 2023)

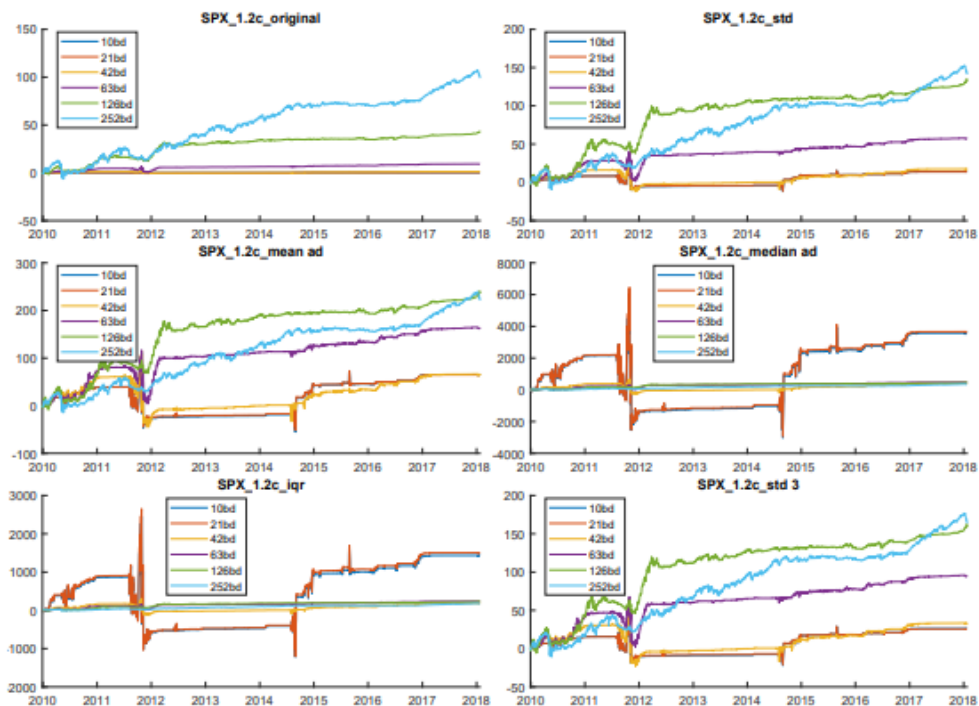


Figure 8: Performance of delta-hedged option strategies across different strikes and maturities

Bormetti et al., (2021) conduct an empirical investigation of trading techniques based on the systematic selling of delta-hedged options with the goal of capturing the so-called volatility risk premium in their article. The performance of various strikes and maturities is compared, and the performance drivers are broken down. Although this study does not directly address the problem of rough gaps in swing and intraday trading, it does provide thorough insights on the behaviour of the S&P 500 index options market. The study examines the volatility risk premium of the S&P 500 index and provides information on the success of delta-hedged option strategies. The findings might be used to influence trading strategies and could be tangentially connected to the present issue.

2.5. Using Gap Trading to Increase Returns: A Concise Approach

Authors Ta, Liu, and Tadesse (2020), said that for both institutional and ordinary investors, the practice of gap trading, a technique focused on spotting and profiting from price discrepancies in the financial markets, has the potential to increase investment returns. Investors can successfully negotiate the changing terrain of financial markets by taking advantage of the patterns and subtleties of gap behaviour. The advice provided in this article will help investors use gap trading tactics to boost their investment profits. Investors looking to maximize their earnings always search for efficient trading tactics in the realm of financial markets. Gap trading is one of the tactics that has drawn interest; this method is focused on spotting and profiting from price gaps that happen between successive trading sessions. Astute investors can take advantage of these voids, which are frequently caused by important news events or changes in market sentiment. A complex strategy that combines knowledge, analysis, risk management, and flexibility is essential to maximize the potential of gap trading and increase investment returns. A firm understanding of gap kinds and patterns is necessary before beginning a gap trading adventure. The characteristics of each of these gaps' common gaps, breakaway gaps, continuation gaps, and fatigue

gaps reflect a different market emotion and its ramifications. Decoding the market's language means being proficient at spotting various gap kinds and determining their relevance. Where author Patton, and Weller (2020) define that the investors may better negotiate the challenging landscape of entry and exit decisions by recognizing gap patterns and associated behavioural clues, improving their accuracy and knowledge of the situation. Comprehensive analysis is key to effective gap trading. A strategic fusion of technical indicators, chart patterns, and fundamental data is essential for supporting trading decisions and goes beyond just identifying gaps. This all-encompassing method of analysis acts as a filter, improving signals and reducing the likelihood of erroneous findings. Investors may improve their ability to forecast price changes accurately by combining gap research with well-established analytical techniques, giving them a solid approach that withstands market volatility (Patton, and Weller, 2020).

According to the authors Alsaad et al., (2018), the trading volume that precedes a gap creation is a revealing sign of the gap's probable success. The expected influence of the gap on pricing has more relevance when there is increased trading volume during the gap. The increase in trading volume reflects the market's participants' active participation and enthusiasm for the price movement. Volume analysis integration into gap trading method develops into a potent instrument that fosters confidence in trading. Because of the additional layer of substantiation provided by volume, which allows investors to act with conviction supported by actual market participation. The early hours of the financial markets are inextricably linked to gap trading. Focusing on gaps that appear during active market hours, when liquidity is at its highest, frequently yields the best outcomes. Due to the increased trading activity at these times, gaps that appear have a higher level of credibility. By minimizing the possibility of rapid reversals, this phenomenon enables trading decisions to reflect real trends and attitudes. Effective gap trading relies on the establishment of precise and

unambiguous entry and exit criteria. The gap type, direction, amplitude, and supporting signs should all be considered while determining these criteria. Disciplined risk management requires that clear profit-taking and stop-loss limits be established. Investors protect themselves against irrational decision-making that might endanger prospective benefits by laying out a well-structured plan. Backtesting a gap trading technique against historical data is a wise move before real-world use. Investors may evaluate how the approach would have performed in different market scenarios using this simulation. Backtesting reveals the strategy's advantages and disadvantages, allowing for adjustments and advancements that increase its efficacy. Due to market volatility, gap trading has inherent risks just like any other trading method. Protecting assets requires effective risk management. The effect of prospective losses can be lessened by using strategies including sensible position size, careful use of stop-loss orders, and portfolio diversification, which can also help to preserve cash for future possibilities. The effectiveness of gap trading depends heavily on being informed because news releases and events frequently cause gaps. Investors are better prepared to foresee possible gaps in the market and act swiftly in response when they are aware of economic data, business earnings reports, and worldwide happenings. This watchfulness synchronizes trading choices with the present market dynamics (Alsad et al., 2018).

Similarly, according to the author Vogl, M. and Roetzel (2022), the financial markets' dynamic demands adaptation. An effective gap trading approach in one market condition may require adjustments in another. Investors may assure the strategy's continuous effectiveness by regularly reviewing its performance and its alignment with changing market circumstances. Gap trading's effectiveness can be increased by combining it with other complementing tactics. Combining gap trading with other techniques, such as trend-following or mean-reversion, might result in a more robust and all-encompassing method of seizing market chances. Discipline and patience are

essential components of effective gap trading. Not all gaps are high probability setups, and seizing every chance might not produce the best outcomes. Successful implementation depends on concentrating on well considered, high-potential settings and maintaining consistency with the plan of action. The financial markets are known for their constant fluctuation. Investors must make a commitment to lifelong learning if they want to make sure a gap trading method stays applicable and efficient. By continuing their education, investors may position themselves to be agile enough to move across the changing terrain while being aware of new trends, methodologies, and market dynamics. By taking advantage of price discrepancies in the financial markets, investors have a sophisticated potential to increase their returns through gap trading. Through a mix of analytical rigor, risk management, flexibility, and continual learning, this strategy's promise may be completely realized (Vogl, and Roe tzel, 2022).

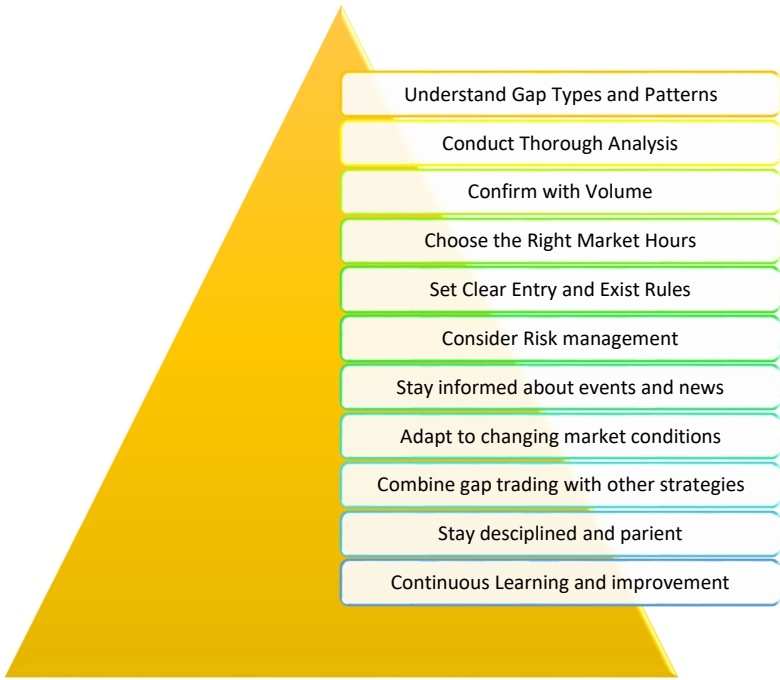


Figure 9: Trading Strategies

(Source: Barchart, 2023)

2.5.1. Psychological Factors in Gap Trading

Psychological factors play a crucial role in the business and influence both decision-making processes as well as results. The transition from tech-driven market research to self-business psychoanalysis has already been acknowledged as crucial for the success of businesses (Kuchta, 2023). Important psychological variables studied in this context included overconfidence, stereotyping, shepherding attitudes and availability bias (Nathanson 2019). These traits often result in cognitive biases and weaknesses that generate hindrance to traders' development, which makes it impossible to use marketing skills efficiently (Kuchta 2023). It is acknowledged that fear and corruption as the two powerful factors create economic losses and can affect business choice (Blandin & Anderson, 2023). It is critical for marketers and organizations to understand these psychological challenges in order to enhance overall decision-making, as well as optimize their performance on the market (Kuchta, 2023). Other subjects addressed in this field include the effect of psychological components associated with stock option exercise and trading behaviour by novice traders.

While there are several scholarly sources listed in this regard, the ones discussed do not directly address conceptual variables that drive considerable disparities towards technological research. Nevertheless, the sources analyse how certain psychological aspects affect trading and market research. As for the paper entitled "Technical research gives you bravery but not money," it touches on psychological factors driving technical analysis in financial markets (Kubińska et al., 2018). In the article entitled "The Effects of Technical Analysis and Psychological Elements on Retail Investors' Business Models," an investigation is conducted as to why investors choose some

issues, looking through them from various angles such as technical analysis and psychological elements (Kumari Kanth MD et al., 2021).

Attitudinal change has a large effect on performance, perceived in terms of decision-making processes and overall outcomes. Since these factors have a negative effect on decision-making and placing businesses in the marketplace, it is crucial that they are understood and addressed. The fight against psychological biases and flaws is one of the most important elements in trading strategies; also, fear or greed must be recognized for rationale to make rational trade decisions.

2.5.2. The Role of Global Economic Events in Gap Trading

The role of macroeconomic events on the global stage in gap trading presents a multidimensional and complex topic. Gap trading strategies are driven by global macroeconomic occurrences that affect market behaviour and present exciting opportunities to traders.

An important fact to remember is the impact of supply chain disruptions on international change and business output, as stressed with the help of research conducted by means of European Central Bank. These disturbances, regularly instigated by means of international occurrences together with the COVID-19 pandemic, have resulted in longer transport intervals in a couple of industries, particularly in evolved nations. Consequently, this influences commercial production and exchange, often main to the emergence of marketplace discrepancies. During the pandemic, there has been a vast upward push in delivery times for technical gadget and equipment, which shows a critical scarcity of intermediate products in those sectors. The supply chain disruptions had a domino effect, resulting in unfavourable effects for international industrial production and trade at the same time as simultaneously riding up inflation (Attinasi et al., 2022).

Economic information is important in influencing gap trading strategies. Indicators inclusive of GDP boom, inflation costs, unemployment figures, and interest fees have

a good-sized effect on marketplace movement. Elevated inflation quotes, consisting of those skilled in a financial system, can also indicate a market this is favourable for traders, at the same time as multiplied unemployment quotes may additionally indicate the opposite. These indicators offer precious insights into the overall state of the economic system, influencing the performance of character equities and broader marketplace movements. Traders can utilise those indicators to broaden customised strategies that are tailored to diverse marketplace conditions, thereby enhancing their decision-making abilities and potential for profit. AI-powered tools along with Easy Gap assist buyers in incorporating indicators into their buying and selling methods by way of analysing gaps and ancient records (Sykes, 2023). Comprehending the correlation amongst worldwide monetary occurrences, monetary indicators, and gap buying and selling is vital for investors. It empowers people to handle market swings with more effectiveness, make nicely knowledgeable choices, and seize possibilities that arise from those dynamic situations.

2.5.3. Emerging Trends and Future Outlook

Technological advances and changing behaviours of market participants have had a significant impact on cross-border trading in recent years. One important factor is the growing impact of social media on investor behaviour and market dynamics. Social media platforms such as Twitter, Reddit, financial forums and others play an important role in influencing market sentiment, often leading to large and sudden price fluctuations that create differences (gap) in the market.

Gap trading is an investment strategy that takes advantage of price differences in the market due to factors such as economic indicators, earnings reports, or major events (Sykes, 2023) This strategy requires more accurately, make faster decisions, and have

a better understanding of market fluctuations. Interval traders attempt to exploit these intervals by predicting price increases or decreases from the interval (Sykes, 2023).

Social media has been highly influential in shaping market sentiment and influencing investment behaviour. Recent research in Chinese banks showed that the spread of rumours through social media channels can have a significant impact on the conversion rate received in the banking market (Zhang et al., 2022) The power of social media rumours influence prices quickly and significantly, thereby in the market The difference is observed (Zhang et al., 2022).

2.5.3.1. Social Media as a Market Driver

Social media has heavily influenced Market sentiments and created imbalance in prices due to its availability of market sensible information for the general public social platform is making Markets communications fast by hastening deployment of ideas, insights perspectives among private Investors. It may cause a rapid response of many traders, when people or market industrialists post useful information or their opinions on the social networks. This leads to wild price swings after market closing or before and when opened, thus resulting in differences.

On the other hand, market volatility (also known as price volatility) can be observed in all types of financial markets such stocks indices commodities and currencies. These intermediaries can be swayed by such economic indicators, earnings announcements, significant news events and announced political and geopolitical changes as well as commercial activities during nights or weekends (Penev 2023). Studies have considered the role of social media in stock market volatility and found that comments from social media can lead to suddenly High price fluctuation ((Zhang et al., 202!)).

Gap trading refers to a type of trade that takes advantage of market price differences based on many variables. This entails precision, speed in making decisions and a thorough understanding of market dynamics (Sykes, 2023). Although intermediate

trading is a strong option, it should be understood that this one of many other avenues. Different business models can be beneficial in identifying different market situations and getting the best result.

2.5.3.2. Case Studies: Social Media Induced Gaps

The recent events have indicated that the attitudes on social media can affect gap trading significantly. For instance, posts (tweet) by influential people or coordinated movement of groups on sites such as Reddit have resulted in huge disparities regarding the shares price some stocks and major market indices (Li et al., 2019). One of the easiest, yet most relevant sources to determine market sentiment is social media data because it gives timely and rich information about public interests (Guan et al., 2021). SMD trading is widely accepted, and expressive trading, a type of SMD trade is considered to have increased stock prices (Reiners, 2021). Businesses benefited immensely from the globalization waves of social media like creating awareness on their products or services as well as networking with individuals worldwide (Alaniz 2022). The current study examines the link between security pricing and investor sentiment based on stock market participants social media postings and applying natural language processing methods (Liu et al., 2022). These activities demonstrate the increasing effects of views expressed through social medias on marketing opportunities and concerns.

Businesses can benefit in various ways from social media globalization, including promoting their products and services and connecting with people from all over the world (Alaniz, 2022). This paper investigates the synergistic pattern between stock prices and investor sentiment using social media messages from stock market investors and natural language processing techniques (Liu et al., 2022). These events underscore the growing influence of social media sentiment in creating trading opportunities and challenges.

2.5.3.3. Algorithmic trading and social media

Integrating data from social media platforms into algorithmic trading systems is a trend that is still emerging. Algorithms are becoming increasingly sophisticated to analyse and react quickly to social media sentiment to identify potential opportunities for difference trading, based on patterns found on social media. This helps create a dynamic market environment and more integrated than it would otherwise be.

Even though surveys, news reports, and financial reports are the primary sources of information used in analysing market sentiment throughout financial history, social media has introduced an important and influential new medium for measurement market perspective. Text analysis is used by these algorithms to identify sentiment, identify trends, and predict future market trends. This is done through the collective knowledge of social media users. According to Wilson (n.d.), social media sentiment can be associated with more effort and lightning speed distribution, although it requires more data filtering. According to He, Lin, and Wang (2021), several machine learning models have now put it through their paces to determine whether or not online social sentiment data can be incorporated into algorithmic trading strategies.

The use of social media in algorithmic trading shows great potential, but it also presents many challenges. These challenges include large amounts of data requiring sophisticated processing and storage, as well as the need to guarantee the accuracy and reliability of the sentiment analysis algorithm. Continued improvements in social media platforms will expand their influence in algorithmic trading channels.

2.5.3.4. Future Outlook and Challenges

The relationship between gap trading and social media is expected to further strengthen in the coming years, possibly increasing opportunities and threats. According to Dwivedi et al., (2021), integrating social media into business processes presents additional challenges. This is even though businesses can reap significant

benefits from social media, such as reduced costs, increased brand awareness, and increased sales. Outside the crowd, market fluctuations can be massive and unforced (Appel et al., 2020). Information shared on social media is not always reliable, and the following of the crowd can influence the volatility of the market. As a result, traders and investors will need sophisticated strategies that can quickly distinguish the noise from the information that drives the market itself. Furthermore, market authorities can take action if they sense that social media is being manipulated (Kolowich, 2022).

The way in which business opportunities are identified and exploited in financial markets is expected to change dramatically in the future, and it is believed that social media marketing will dictate the future. According to Appel et al.'s (2020) prediction, the increasing influence of social media on algorithmic trading platforms will create a more connected market environment. On the other hand, it may be difficult for social media companies to avoid providing misleading information and market manipulation due to the lack of clear regulations and penalties (Anderson, Rainie, and Vogels, 2021). Such a situation can be difficult to deal with.

Ultimately social media influence on investor behaviour as well as cross market trading brings important alternatives in market dynamics. Organizations that adjust to this evolving environment as they pursue social media analytics in their operations can understand the opportunities available through understanding breakdowns by knowing risks. The key to success in gap business viability lies through keeping up to date with new market trends. The traditional market sentiment analysis has been based on surveys, news reports and financial statements. On the other hand, social media has created significant new elements for sentiment analysis of markets. These algorithms obtain collective intelligence from media users with the help of text analytics which enables them to understand sentiment, detect trends and predict future market tendencies. On the other hand, several recent studies produce solid evidence that suggest investors expressing their opinions on social media sites may have a bearing

on financial markets (Li et al., 2019). Integrating social media data into algorithmic marketing programs is a growing trend. Algorithms are increasingly developing to test social media sentiment and respond quickly to explore potential opportunities for intertrade based on social media actions. This creates a dynamic and highly interconnected market environment (He, Lin, & Wang, 2021).

However, the absence of appropriate laws and clear punishments for social media businesses, on the other hand, gives upward push to a problem given that it can lead to the dissemination of deceptive facts and the manipulation of the market. To ensure that the integrity of the monetary markets is preserved, it's miles vital for regulatory bodies to handle the problems which have been presented.

Chapter 3

3. Research Methodology

Financial markets frequently have price gaps, which can allow traders and investors to capitalise on market abnormalities. In this paper, we'll outline a system for examining price gaps and creating trading plans that profit from them. We can take the following actions:

From Investing.com we can examine index S&P 500 first 15 minutes chart. We selected the S&P 500 because it is a well-known index that measures the performance of 500 large-cap U.S. businesses operating in a range of industries. It is regarded as one of the most accurate gauges for the American stock market.

Additionally, the S&P 500's constituents often have large trading volumes and liquidity. This indicates that there is significant market activity during the opening hours and that the index is actively traded. Tighter bid-ask spreads, which lessen the effect of slippage and provide better execution while trading or analysing opening gaps, are often the outcome of more liquidity and volume. We chose a 15-minute time limit because we want to be able to catch the quick price changes that happen as soon as the market opens. These brief price changes might reveal important details regarding the size and shape of the emerging gap. There are many kinds of gaps, but there are only two possible directions for them: up and down.

Second, this study will include a sample of data from the previous three years (2020, 2021 & 2022). We will do research on 750 samples, or around 250 samples every year. The reason we mention samples is because it's important to analyse gaps and comprehend how gaps behave. The outcome will be precise regardless of how many data we analyse.

3.1. Using a pre-experimental approach

We will analyse the opening gaps in three different ways:

For us, it will be helpful to analyse the price gaps depending on their magnitude and direction. A wide gap might point to a substantial shift in market sentiment, whilst a small gap can point to a more subtle shift. The gap's direction might be important as well since it can reveal whether the market is bullish or bearish. In further studies, we will thoroughly explore the effect that trade volume and gap size have on trading success.

In a second method, we will analyse the data based on the time needed to fill in the gaps. Some opening voids are closed the same day, while others take longer. We will determine the size of the gaps that are filled on the same day and the time that the gaps are not filled. The S&P 500 data from the previous three years will be used for every analysis. Traders can determine what position traders can take after a gap by examining the size of the gap, the length of time it takes to close the gap, or if it continues along the gap's trend.

The research methodology for analysing the gaps during the provided period and studying their filling on the same day involves several key steps. Firstly, relevant data including historical price data, trading volume data, and information on price gaps will be collected for the specified period. The gaps will then be analyzed and categorized based on their size, with a focus on gaps that are 0.5% above the previous closing price.

Next, the study will determine the percentage of gaps in each size category that are filled on the same day by averaging the data. This analysis will provide insights into whether gaps of the specified size tend to be filled promptly or not.

In addition, a linear regression analysis will be conducted to explore the relationship between gap size and trading volume. Gap size will serve as the independent variable, while trading volume will be the dependent variable. By examining this relationship, the study aims to ascertain whether trading volume can be utilized as a predictor for gap filling or the continuation of gap direction.

The findings from both the analysis of gap filling on the same day and the linear regression analysis will be interpreted and discussed in relation to the research objective. The significance of the relationship between gap size and trading volume will be examined, shedding light on the potential use of trading volume as a tool for predicting gap behaviour and guiding trading strategies.

Chapter 4

4. Finding and Discussion

The study aimed at looking into the effect of diverse elements on opening gaps stock market behavior. This section then evaluates the findings related to a hypothesis and analyzes them from the perspective of those outcomes obtained in conjunction with data analysis. The principal hypotheses, however, were regarding the impact of size of gaps on market reversal, gap size relationship to trading volume and economic variables such as gold prices and the value of USD.

Finding related to hypothesis 1 (Impact of Gap Magnitude on Market Reversal)

The hypothesis posits that the magnitude of the opening gap influences the likelihood of market reversal.

Data Observation

Gap Category		Average Gap % Change		Average Post Gap Change	Direction	
Gaps > 2%		2.97%		-1.46%		
Gaps 1% - 2%		1.39%		-0.79%		
Gaps 0.5% - 1%		+0.70%		-0.33%		
Gaps -0.5% to 1%		-0.70%		+0.18%		
Gaps -1% to 2%		-1.37%		+0.70%		
Gaps < -2%		-3.39%		+2.09%		
Note: Detail attached in appendix						
Regression Statistics						
R	0.991901913					
R Square	0.983869406					
Adj. R Sqr	0.979836757					
Std Error	0.003179937					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	1	0.002467	0.002467	243.976	9.181E-05	
Residual	4	4.04E-05	1.01E-05			
Total	5	0.002508				
	Coefficient	Std Error	T stat	P-value	Lower 95%	Upper 95%
Intercept	0.00049457	0.0013	0.380346	0.723024	-0.00312	0.004105
Average Post Gap Direction Change	-1.78652608	0.114376	-15.6197	9.81e-05	-2.10409	-1.46897

Interpretation

- i. Gaps > 2%: Large positive gaps are followed by a notable average reversal of -1.46%

This suggests that extremely positive openings often lead to subsequent downward corrections, possibly due to initial over-optimism or overreaction in the market.

- ii. Gaps 1% - 2%: Moderate positive gaps show a smaller average reversal of -0.79%.

The trend of reversal continues but is less pronounced, indicating that while positive sentiment is corrected, it is not as extreme as in larger gaps.

- iii. Gaps 0.5% - 1%: Smaller positive gaps result in an even lesser reversal trend, averaging -0.33%.

This shows a diminishing reversal trend as the gap size decreases, suggesting that smaller upward gaps are often followed by slight downward adjustments.

- iv. Gaps -0.5% to 1%: Small negative gaps have an average post-gap change of +0.18%.

In contrast to positive gaps, small negative gaps are often slightly corrected upwards, though the magnitude of reversal is modest.

- v. Gaps -1% to 2%: Moderate negative gaps lead to a more significant average reversal of +0.70%.

These findings imply that larger negative openings are more likely to be met with a bullish response, perhaps due to market perceptions of overselling or an overreaction to negative news.

- vi. Gaps < -2%: The largest negative gaps show the most significant average reversal of +2.09%.

This significant reversal may indicate that extreme negative gaps are often perceived as excessive bearishness, leading to a strong bullish counter-reaction as the market seeks equilibrium.

Discussion

1. Reversal Trend:

Positive Gaps: For positive gaps (greater than 0.5%), a trend of reversal is observed, with the magnitude of the reversal being inversely proportional to the size of the gap. Larger positive gaps tend to be followed by more significant reversals.

Negative Gaps: For negative gaps, the reversal trend is also evident. Notably, the larger the negative gap, the stronger the reversal, as seen in gaps below -2%.

2. Magnitude Influence

The data clearly shows that the magnitude of the gap significantly influences the direction and extent of the post-gap market movement. This is true for both positive and negative gaps, suggesting that larger deviations from the previous closing price are more likely to induce corrective market responses.

3. Market Psychology

Overreaction Adjustment: The reversal trends, especially in large gaps, could be attributed to market overreactions. Large gaps might cause an initial emotional response from traders, which is then corrected as more rational assessments prevail.

Perception of Value: In the case of large negative gaps, the market might perceive an oversold condition, leading to buying opportunities and a subsequent positive reversal.

4. Liquidity and Volatility

Volatility Impact: Larger gaps, particularly those beyond 2% (positive or negative), could be indicative of higher market volatility. High volatility often leads to rapid price changes and may contribute to the observed reversal trends.

Liquidity Concerns: Gaps may also be influenced by liquidity. For example, lower liquidity could result in larger gaps due to less efficient price discovery, which then adjusts as liquidity improves.

5. Statistical Significance

Regression Analysis Results: The regression analysis shows a high Multiple R (0.991901913) and a high R Square (0.983869406), indicating a very strong linear relationship.

P-Value: The extremely low P-value (9.8103E-05) for the regression indicates that the relationship between gap magnitude and market reversal is statistically significant.

Coefficients: The coefficient for the average post-gap direction change is -1.786526084 with a very low P-value, confirming its significance in predicting the market's response.

The analysis indicates a strong and statistically significant relationship between the magnitude of opening gaps and subsequent market reversals. Larger gaps, whether positive or negative, are more likely to be followed by significant market corrections. This trend aligns with market psychology, where substantial initial movements, possibly driven by emotional or reactionary trading, are often counterbalanced by subsequent market activity. The high R Square value and the low P-value in the regression analysis underscore the robustness of this relationship. However, it is essential to consider other external factors like news events, economic indicators, and broader market trends that might also influence market movements post-gap.

Finding related to:

Hypothesis 2: Correlation between Gap Size and Trading Volume

Hypothesis 3: Influence of Economics Indicators (Gold & USD)

Correlation Analysis	After Gap Direction from Open	Volume	Gold	USD
After Gap Direction from Open	1			
Volume	0.060470861	1		
Gold	-0.00458726	-0.081384674	1	
USD	-0.0039142	0.001085825	0.00708537	1

Understanding the Correlation in the context of Hypothesis 2

1. After Gap Direction from Open (0.060470861)

This correlation coefficient indicates a positive, but very weak relationship between the direction after a gap from the opening price in a trading session and the trading volume. It suggests that there might be a slight tendency for larger gap sizes to correlate with higher trading volumes, but the relationship is not strong.

2. After Gap Direction from Open and Gold (-0.00458726)

This is a negative correlation but it's extremely weak, almost negligible. It implies that the direction after the gap from the opening price has virtually no linear relationship with the price of gold.

3. After Gap Direction from Open and USD (-0.0039142)

Like the correlation with gold, this is a negative and very weak correlation with the value of USD. It indicates that the after-gap direction from the open has almost no linear relationship with the value of USD.

4. Volume and Gold (-0.081384674)

This is a weak negative correlation. It suggests that there is a slight tendency for trading volume to decrease as the price of gold increases, but the relationship is not strong.

5. Volume and USD (0.001085825)

This correlation is positive but extremely weak, almost negligible. It indicates that there is no significant linear relationship between trading volume and the value of USD.

6. Gold and USD (0.00708537)

This is a positive but very weak correlation. It implies that there is no significant linear relationship between the price of gold and the value of USD.

Discussion

The correlation coefficient with trading volume is 0.060470861, indicating a very weak positive correlation. This suggests that while there may be a slight trend where larger gap sizes (either positive or negative) from the open could be associated with higher trading volumes, this relationship is not strong enough to be considered significant in most analytical contexts. It's crucial to remember that correlation does not imply causation. Even if there's a correlation, it doesn't necessarily mean that the gap size causes the change in trading volume.

Understanding the Correlation in the context of Hypothesis 3

1. After Gap Direction from Open (0.060470861)

This indicates a very weak positive correlation. Changes in the direction of stock prices after opening (compared to the previous day's closing) seem to have a very slight positive relationship with trading volume. However, the correlation is so weak that it might not be significant for practical purposes.

2. After Gap Direction from Open and Gold (-0.00458726)

This is a negligible negative correlation. Essentially, it suggests there's no meaningful relationship between the direction of stock prices after the market opens and the price of gold.

3. After Gap Direction from Open and USD (-0.0039142)

Like the relationship with gold, this is a negligible negative correlation with the value of the USD. It implies that the direction of stock prices after opening has almost no relationship with the value of the US dollar.

4. Volume and Gold (-0.081384674)

This indicates a weak negative correlation. As trading volume increases, the price of gold might decrease slightly, but the relationship is not strong.

5. Volume and USD (0.001085825)

There is essentially no correlation between trading volume and the value of the USD.

6. Gold and USD (0.00708537)

This shows an extremely weak positive correlation. The relationship between the price of gold and the value of the USD is almost negligible.

Discussion

Gold: Gold shows very weak to negligible correlations with all the other variables. This implies that as an economic indicator in this context, gold does not significantly influence the direction of stock prices after opening, trading volume, or the value of the USD.

USD: Similarly, the USD shows negligible correlations with the other variables. This indicates that the value of the USD has almost no influence on the direction of stock prices after opening, trading volume, or the price of gold in this dataset

Regression Statistics						
R	0.060602779					
R Square	0.003672697					
Adj. R Sqr.	0.000686701					
Std Error	0.101070859					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	3	0.000423	0.000141	1.229973	0.297551863	
Residual	1001					
Total	1004					
	Coefficient	Std Error	T stat	P-value	Lower 95%	Upper 95%
Intercept	-0.0024	0.00796	-0.30390	0.761261	-0.018051	0.01320
Volume	6.09435	3.18834	1.91144	0.056232	-1.62244	1.23509
Gold	3.37552	2.92636	0.01153	0.990799	-5.70876	5.77627
USD	-7.1753	5.68433	-0.12622	0.899576	-0.000118	0.00010

Discussion on Hypothesis 2

Variable	P-value	Interpretation Threshold
Trading Volume	0.056232	> 0.05 (Not Significant)

Statistical Significance:

Observation: The P-value for the trading volume is 0.056232113.

Interpretation: In statistical analysis, a P-value less than 0.05 is typically considered significant. Since the P-value here is slightly above 0.05, it indicates that the evidence against the null hypothesis is not strong enough to be conclusive.

Correlation Strength:

Observation: A P-value close to the threshold suggests a potential but not definitive relationship.

Interpretation: While there may be a trend or slight correlation between gap size and trading volume, it's not statistically strong enough to assert with high confidence. This could imply that factors other than just the gap size influence the trading volume.

Market Dynamics:

The stock market is influenced by a multitude of factors including investor sentiment, economic indicators, news, and events. Trading volume might be influenced by these factors as much as, or even more than, by the size of the opening gap.

Data Considerations:

The reliability of the result can be affected by the data sample size, the range of the data (such as specific market conditions or periods), and the method of data collection. A larger or more diverse dataset might yield different results.

Based on the regression analysis, there is no strong statistical evidence to support the hypothesis that there is a significant correlation between the size of the opening gap and the trading volume. While the P-value is close to the threshold, it exceeds the conventional limit for statistical significance. This suggests that, in this analysis, the trading volume may not be reliably predicted based on the size of the opening gap alone.

Discussion on Hypothesis 3

Economic Indicator	P-value	Interpretation Threshold
Gold Prices	0.990799	> 0.05 (Not Significant)
USD Value	0.899576	> 0.05 (Not Significant)

Statistical Significance:

Observation: Both gold prices and USD value have high P-values, far exceeding the conventional significance threshold of 0.05.

Interpretation: The high p-value means that there is insufficient statistical evidence to prove the hypothesis that these monetary variables have a strong impact on differential openings in stock markets.

It is possible to interpret economic indicators, including the US dollar prices and gold price, taken as indicators of conditions from economy as whole but this study suggests that their direct effects on specific behaviors in gap widening might be weakened.

Stock price is influenced by several factors including financial data. Since gap openings are so intricate, market trends cannot be determined based on one or several factors.

Data Scope and Limitations:

It is important to address the size of the data used in the study. Different results will occur when there are changes in market conditions or times. Furthermore, some important variables or their interactions may have been omitted by the model.

Practical Implications:

Traders and analysts must keep in mind that predicting opening gaps using solely gold price movements or variations of the value of USD will not always result in a successful prediction. An integrated approach that considers the economic, financial and geopolitical aspects would probably do a better job.

Regression research indicates that economic indicators of gold price and the value USD do not seem to affect the opening gaps in stock market behavior with any significant relevance. But high p-values evidence for H_0 and point out that these indicators are not strong predictors here. It is hence assured that factors beyond only critical economic indicators cause stock market behavior, especially those opening gaps.

Chapter 5

5. Conclusion

Price differentials within the financial markets are the subject of numerous studies regarding their effects on profitable trading. The analysis is complete ranging from size and development gaps; it also covers the effect on economic indicators. It contains a crucial evaluation of the results, examination of the study design, limitations section and references for future research.

5.1. Critical analysis of the results

Significance in the literature:

Supporting theoretical propositions through arguments: An important finding in literature, the first finding of the study is that the data provided support for the theoretical propositions. Current economic theories are based on this reconciliation between theory and actual evidence. This means that although these principles are theoretical, they have real-world applications. This part of the study adds to the body of knowledge by making these theoretical models seem more plausible.

Overreaction to Ideological Concepts: Notable is the finding that people overreact to market events in line with broader psychological beliefs. This means that economic indicators are not even the only important determinants of market behaviour. The role is heavily influenced by psychosocial factors including current market trends and psychological biases. Rethinking the role of news and media in influencing market dynamics would have provided us with this important insight into market dynamics. This highlights the importance of behavioural economics in studying the impact of psychology on investment decisions.

The presence of modest connections between the movement of stock prices after the market opens and both trading volume and economic indicators like gold and USD, is specifically charming. This means that the first fluctuations in stock prices are less impacted by way of these factors than conventionally believed. This finding has the

potential to essentially alter our comprehension of market dynamics during the critical moments following the opening of the market. Furthermore, it may have a great impact on buying and selling techniques that depend upon these interconnections.

Economic Indicators and Trading Volume: The weak correlation between the trade volume and gold, and the insignificant correlation between the trade volume and the USD, undermine the traditional assumption that these currency variables significantly affect changes in the trade volume. Thus this finding suggests that there may be other, less recognized factors that may have a greater effect on sales volume. It provides a new opportunity to examine the underlying factors affecting trading volume in the stock market.

Implications for traders and market strategists: These findings suggest that traders and market analysts should reconsider their reliance on traditional financial indicators for predicting market performance. This implies a potential shift towards a broader approach that covers a broader range of factors, including psychological and social factors. This has the potential to create more sophisticated and complex marketing strategies and models.

Research findings on gap behaviour provide valuable information that sophisticated traders can use to make more educated decisions. This phenomenon, known as the mismatch between expected theoretical results and actual market results, creates a distinct advantage for traders who have the ability to accurately assess and react to these discrepancies and to see and hear them under subtle differences in these equations. Nevertheless, this perspective emphasizes the fundamental volatility is the uncertainty of financial markets. This highlights the idea that relying solely on traditional indicators such as financial statistics or historical observations cannot always provide accurate predictions of market behaviour. This insight forces a different approach encourage commercial use, including integration of signals. While identifying differential behaviour can be a powerful tool for shrewd traders, it also serves as a

warning against over-reliance on any individual statistics or model Data-driven analysis that combines and discovers the inherent uncertainties in the market make this strategy more effective in the complex form modern financial markets development.

5.2. Justification of findings, recommendations and conclusions based on the data

The results of this study, which rely on quantitative data analysis, prove that there are challenges and opportunities in traditional marketing strategies. The use of regression statistics is particularly good in a quantitative sense, supporting its implications for observed market behaviour. These cases illustrate the limitations of classical trading strategies based on limited signals or prior information. The weak correlation between the difference patterns obtained for stock price, trading volume and general economic indicators suggests that traditional methods failed to capture the nuances of current market behaviour.

Considering these findings, the recommendation to conduct a comprehensive market study is entirely warranted. This approach should cover all possible market segments to give a complete picture of the financial situation. Thus, by incorporating a various view of market dynamics can be introduced by considering multiple aspects such as economic indicators, behavioural insights and psychoanalysis, taking continuous dynamics in emerging markets, acknowledge geopolitical events, technological developments, changes in consumer behaviour and other mediating factors.

A comprehensive approach that considers not only the direct impact of traditional indicators but also the indirect and subtle explanations associated with intermediary factors may enable market players to better understand such holistic systems. This concept is linked to the changes in the financial markets today, there are many factors besides conventional economic and financial indicators that play an important role in

it. A multifaceted approach that marketers can take will allow them to better understand and navigate this sophisticated and often unpredictable economy of today.

5.3. Critique by research methods and findings

The methodology adopted in this study makes it possible to understand the complexity of market dynamics through a comprehensive analysis of the intermediary market. By learning everything about the middle market and all its derivatives, we can learn everything we need to know. This level is necessary to appreciate the subtle influence of the intermediary market on market behaviour and economic mechanisms.

However, there are some drawbacks to using such historical data in such studies. Are the dangers of selection bias and the practice of over-intervention. This bias is evident when the data used for analysis are not fully representative of the market as a whole or when certain interpretations of the data dominate the interpretation and conclusions of the results ultimately but there are many factors that determine financial markets. Some of these variables are also unstable and change regularly while others have no discernible pattern. However, there are many obstacles to a model that relies entirely on historical data to predict market behavior in such a dynamic environment.

One way to overcome these limitations is to expand their scope by covering new markets in different contexts and adding a number of economic factors to be considered. This will improve the accuracy and relevance of the analysis. In this way, we can analyze various market trends and identify trends that are common to all markets and unique to some of them. Taking a holistic view of market dynamics also opens the way to understanding how one economic indicator interacts with another in market conditions.

In the end, this study demonstrates that reliance entirely on historic records has great regulations, but its technique is dynamic since it looks at intermediary marketing in element. To overcome those boundaries and boost the precision of the usage of these

results in forecasting future marketplace behaviours, a much broader variety of economic indicators from extraordinary markets should be protected.

5.4. Future Directions

Comprehensive analysis is essential for accurate understanding and forecasting in the complex financial markets. To do this, we must combine metrics that look at the entire economy with advanced machine learning models, accurately analyse global markets, and use behavioural economics techniques rationally. By taking a comprehensive view of all these factors, this study aims to identify their importance in the understanding and behaviour of a market.

Comprehensive economic metrics: Therefore, several financial measures are wished for a comprehensive evaluation of marketplace phenomena. Price and interest rate information in addition to others offer insights on the status of the economic system in standard. Through observing these indicators, analysts can detect patterns and outliers within the market that allows them to understand how unique monetary sectors are affecting each other differently. This extensive view is vital due to the fact of finding the underlying reasons for market volatility.

Machine learning model: The machine learning model contributes to a better understanding of factors influencing market effects on connected web sites. These systems can explore patterns and combinations that can go through human eyes when working with big data. Machine learning can increase accuracy in predicting market trends, spotting trends, assessing risk, and evaluating opportunities by analysts. In today's world of growing and evolving data-driven industries advanced research skills are valuable today.

Evaluation of Foreign Marketplace: It is consequently imperative to include foreign marketplace data evaluation for higher knowledge of the workings of world marketplaces. In global enterprise, one must consider that every market has its political system, economic situation and regulation. These foreign markets offer a

better and prolonged view of world finance due to the fact they illuminate how specific economic environments affect investor behavior and market traits.

Incorporating behavioral finance strategies: Other behavioral finance techniques, which include investor sentiment evaluation, reinforce marketplace research. The modifications inside the marketplace because of mental forces (which isn't like basics) are perhaps more understandable for analysts if they recognize mood and mindset of investors. A higher expertise of the market's dynamics can be provided with the aid of sentiment evaluation that permits predicting how markets will react to some modifications like coverage shifts or information stories and other announcements.

It is a whole-market technique toward market analysis that mixes behavioural finance techniques with worldwide markets, device learning and general economy metrics. This article has therefore proven the importance of each component so that it will provide a entire and detailed photograph of market dynamics. In the destiny, those analytical gear might be capable of evolve and alter with an more and more volatile global economic markets surroundings. Finally, from an empirical point of view, the study contributes a lot about trading in financial markets. Even though it has a few similarities with the present study, this paper also focuses on new issues including empirical analysis of application to foretell market behaviour. Since the limits of traders are well determined in financial markets, recommendations on a more universal and versatile trading philosophy seems to be appropriate. In the future, research can add to these foundations by studying multiple and heterogeneous data sources using advanced analytical methods in order improve understanding of intermediary marketing mechanisms.

References

- Alaniz, M. (2022). Social Media and Globalization. [online] Legacy Marketing. Available at: <https://www.legacymarketing.com/blog/social-media-and-globalization/>.
- Alsaad, A., Mohamad, R., Taamneh, A. and Ismail, N.A., (2018). What drives global B2B e-commerce usage: an analysis of the effect of the complexity of trading system and competition pressure. *Technology Analysis & Strategic Management*, 30(8), pp.980-992.
- Aiche Avishay, Cohen, G. and Griskin Vladimir (2023). Stocks Opening Price Gaps and Adjustments to New Information. doi: <https://doi.org/10.1007/s10614-023-10363-w>.
- Anderson, J., Rainie, L. and Vogels, E. (2021). Experts Say the ‘New Normal’ in 2025 Will Be Far More Tech-Driven, Presenting More Big Challenges. [online] Pew Research Center: Internet, Science & Tech. Available at: <https://www.pewresearch.org/internet/2021/02/18/experts-say-the-new-normal-in-2025-will-be-far-more-tech-driven-presenting-more-big-challenges/>.
- Appel, G., Grewal, L., Hadi, R. and Stephen, A.T. (2020). The Future of Social Media in Marketing. *Journal of the Academy of Marketing Science*, [online] 48(1), pp.79–95. doi:<https://doi.org/10.1007/s11747-019-00695-1>.
- Balcilar, M., Ozdemir, Z.A. and Ozdemir, H., (2021). Dynamic return and volatility spillovers among S&P 500, crude oil, and gold. *International Journal of Finance & Economics*, 26(1), pp.153-170.
- Barchart.com. (2023). *S&P 500 Index Performance Report - Barchart.com*. [online] Available at: [https://www.barchart.com/stocks/quotes/\\$SPX/performance](https://www.barchart.com/stocks/quotes/$SPX/performance) [Accessed 28 Aug. 2023].

Bormetti, G., Caldana, R., and Treccani, M., 2021. The Volatility Risk Premium: An Empirical Study on the S&P 500 Index. *Journal of Derivatives*, 29(3), 1-17.

https://doi.org/10.3905/jod.2021.e1_001

Chan, K. F. and Fong, W. M., 2019. Statistical Arbitrage with Mean-Reverting Overnight Price Gaps on High-Frequency Data of the S&P 500. *Journal of Risk and Financial Management*, 12(2), 51. <https://doi.org/10.3390/jrfm12020051>

CHEN, J. (2021). *Gap Definition*. [online] Investopedia. Available at: <https://www.investopedia.com/terms/g/gap.asp>.

Collins, A. and Halverson, R., (2018). *Rethinking education in the age of technology: The digital revolution and schooling in America*. Teachers College Press.

Doran, J.S., (2020). Volatility as an asset class: Holding VIX in a portfolio. *Journal of Futures Markets*, 40(6), pp.841-859.

Dwivedi, Y.K., Ismagilova, E., Hughes, D.L. and Carlson, J. (2021). Setting the Future of Digital and Social Media Marketing research: Perspectives and Research Propositions. *International Journal of Information Management*, [online] 59(1), pp.1–37. doi:<https://doi.org/10.1016/j.ijinfomgt.2020.102168>.

El Ghorayeb, A., (2021). Trading performance in a financial crisis: momentum and the Covid-19 flash bear market.

Feng, Z., Li, W., Zhang, H. and Zhang, X., (2023). A Framework of a Blockchain-Supported Remanufacturing Trading Platform through Gap Analysis. *Sustainability*, 15(16), p.12120.

Gereffi, G., (2019). Global value chains, development, and emerging economies. *Business and development studies*. Routledge, pp.125-158.

Guan, C., Liu, W. and Cheng, J.Y.-C. (2021). Using Social Media to Predict the Stock Market Crash and Rebound amid the Pandemic: The Digital ‘Haves’ and ‘Have-mores’. *Annals of Data Science*. doi:<https://doi.org/10.1007/s40745-021-00353-w>.

Guo, J., Gu, F., Liu, Y., Liang, X., Mo, J. and Fan, Y., 2020. Assessing the impact of ETS trading profit on emission abatements based on firm-level transactions. *Nature communications*, 11(1),p.2078.

<https://www.tandfonline.com/doi/abs/10.1080/14693062.2020.1785380>

Harris, L. and Gurel, E., 2003. The Price Response to S&P 500 Index Additions and Deletions: Evidence of Asymmetry and a New Explanation. *Journal of Finance*, 58(4), 1823-1844. <https://doi.org/10.1111/1540-6261.00580>

He, C., Lin, M. and Wang, N. (2021). Trading on online social mood: A machine learning strategy based on twitter sentiment. *International Journal of Financial Engineering*, p.2141011. doi:<https://doi.org/10.1142/s2424786321410115>.

In, S.Y., Rook, D. and Monk, A., (2019). Integrating alternative data (also known as ESG data) in investment decision making. *Global Economic Review*, 48(3), pp.237-260.

Johnston, H. and Land-Kazlauskas, C., (2018). Organizing on-demand: Representation, voice, and collective bargaining in the gig economy.

Kolowich, L. (2022). 8 of the Top Marketing Challenges Marketers Will Face This Year [New Data]. [online] Hubspot.com. Available at: <https://blog.hubspot.com/blog/tabid/6307/bid/33820/5-major-challenges-marketers-face-and-how-to-solve-them.aspx>.

Li, D., Wang, Y., Madden, A., Ding, Y., Tang, J., Sun, G.G., Zhang, N. and Zhou, E. (2019). Analyzing stock market trends using social media user moods and social influence. *Journal of the Association for Information Science and Technology*. doi:<https://doi.org/10.1002/asi.24173>.

Liu, Q., Lee, W.-S., Huang, M. and Wu, Q. (2022). Synergy between stock prices and investor sentiment in social media. *Borsa Istanbul Review*. doi:<https://doi.org/10.1016/j.bir.2022.09.006>.

Nti, I.K., Adekoya, A.F. and Weyori, B.A., (2020). A systematic review of fundamental and technical analysis of stock market predictions. *Artificial Intelligence Review*, 53(4), pp.3007-3057.

Patton, A.J. and Weller, B.M., (2020). What you see is not what you get: The costs of trading market anomalies. *Journal of Financial Economics*, 137(2), pp.515-549.

Pedersen, L.H., (2019). *Efficiently inefficient: how smart money invests, and market prices are determined*. Princeton University Press.

Penev, M. (2023). Understanding the Market Gap - when it happens and why? [online] Eightcap Labs. Available at: <https://www.eightcap.com/labs/understanding-the-market-gap-when-it-happens-and-why/>

Ramodike, N.M., (2022). *A comparative analysis of traditional and AI ETFs tracking the S&P 500 By* (Doctoral dissertation, University of Johannesburg).

Reiners, L. (2021). INSIDER TRADING AS A RESPONSE TO SOCIAL-MEDIA-DRIVEN TRADING. [online] The FinReg Blog. Available at: <https://sites.duke.edu/thefinregblog/2021/07/01/insider-trading-as-a-response-to-social-media-driven-trading/>

Rout, B.S., (2018). *Performance of commodity derivatives market in India* (Doctoral dissertation, Department of Banking Technology, Pondicherry University).

- Shankar, S.G., Miller, J.M. and Balakrishnan, P.V. (Sundar) (2020). Evolutionary disruption of S&P 500 trading concentration: An intriguing tale of a financial innovation. *PLOS ONE*, 15(3): <https://doi.org/10.1371/journal.pone.0230393>.
- Siegel, J.J., 2021. *Stocks for the long run: The definitive guide to financial market returns & long-term investment strategies*. McGraw-Hill Education.
- Strawser, J.A., Hechavarría, D.M. and Passerini, K., (2021). Gender and entrepreneurship: Research frameworks, barriers and opportunities for women entrepreneurship worldwide. *Journal of Small Business Management*, 59(sup1), pp.S1-S15.
- Sykes, T. (2023). Gap Trading Strategy: Definition, Examples, and Tips. [online] Available at: <https://www.timothysykes.com/blog/gap-trading-strategy/>.
- Ta, V.D., Liu, C.M. and Tadesse, D.A., (2020). Portfolio optimization-based stock prediction using long-short term memory network in quantitative trading. *Applied Sciences*, 10(2), p.437.
- Tushar, W., Saha, T.K., Yuen, C., Morstyn, T., McCulloch, M.D., Poor, H.V. and Wood, K.L., 2019. A motivational game-theoretic approach for peer-to-peer energy trading in the smart grid. *Applied energy*, 243, pp.10-20. <https://www.sciencedirect.com/science/article/pii/S0306261919305185>
- Vogl, M. and Roetzel, P.G., (2022). Chaoticity versus stochasticity in financial markets: Are daily S&P 500 return dynamics chaotic?. *Communications in Nonlinear Science and Numerical Simulation*, 108, p.106218.
- Vos, R., Martin, W. and Resnick, D., (2023). *The political economy of reforming agricultural support policies* (Vol. 2163). Intl Food Policy Res Inst.

Walker, T.A., 2021. *Trading places*. Covenant Books, Inc..
https://books.google.com/books?hl=en&lr=&id=FkdFEAAQBAJ&oi=fnd&pg=PT9&dq=Opening+gaps+helps+to+surge+returns+in+swing+and+intraday+trading.&ots=TQwXM2N7K4&sig=-WVP10cSCedTxcPbkUOrP_bT64

Wilson, I. (n.d.). News Sentiment vs. Social Media Sentiment in Algorithmic Trading. [online] Available at:
<https://maherou.github.io/Teaching/files/CS373/SamplePapers/2022-Fall-Wilson.pdf>

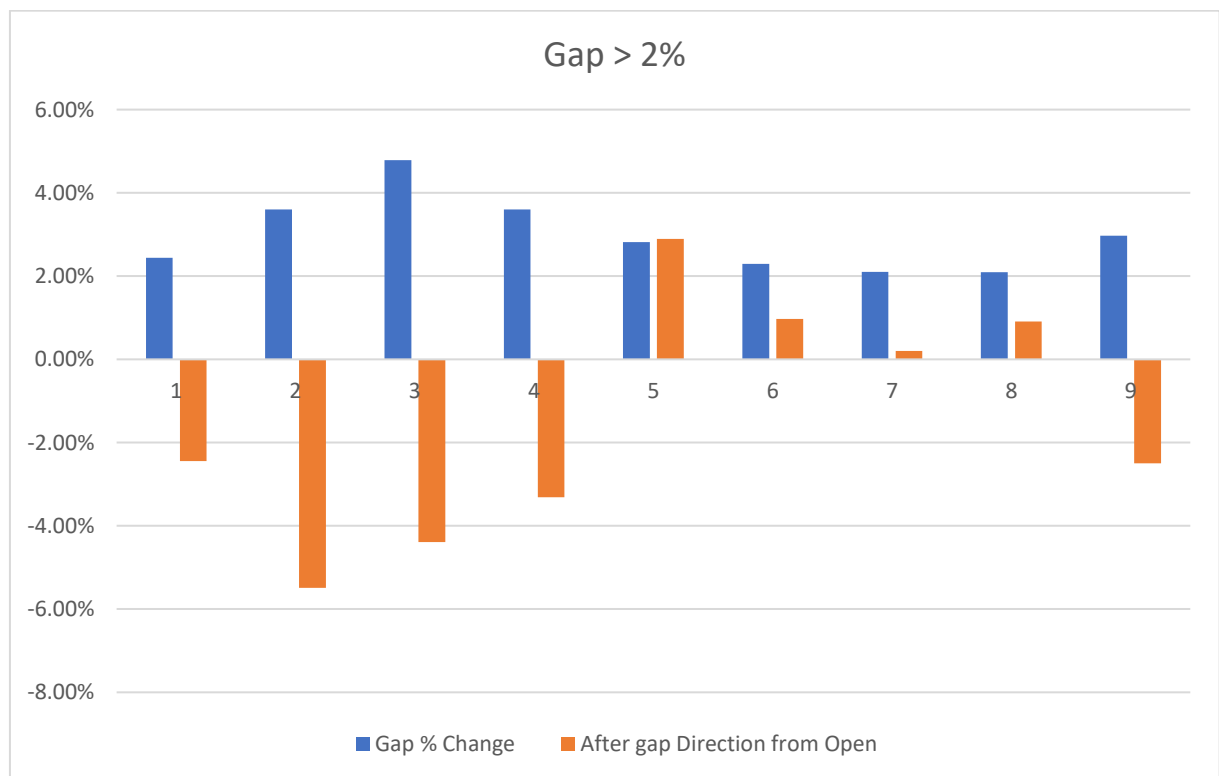
Ycharts.com. (2023). *S&P 500 3 Year Return*. [online] Available at:
https://ycharts.com/indicators/sp_500_3_year_return [Accessed 28 Aug. 2023].

Zarattini, C. and Aziz, A., 2023. Can Day Trading Really Be Profitable? Evidence of Sustainable Long-term Profits from Opening Range Breakout (ORB) Day Trading Strategy vs. Benchmark in the US Stock Market. *Evidence of Sustainable Long-term Profits from Opening Range Breakout (ORB) Day Trading Strategy vs. Benchmark in the US Stock Market* (April 11, 2023).
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4416622

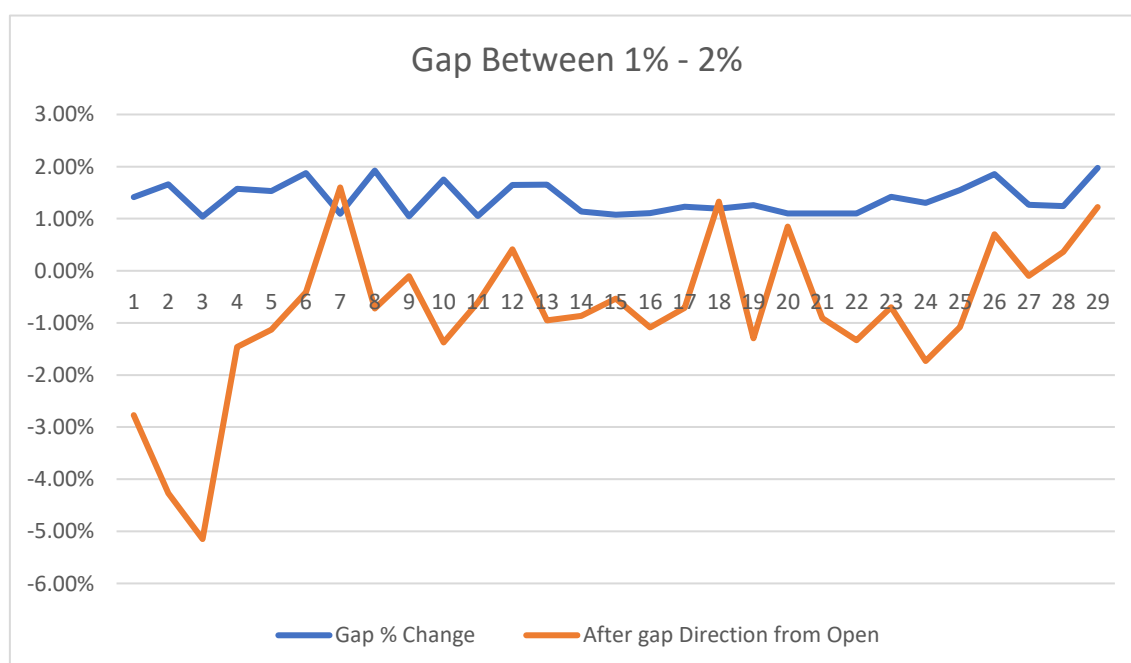
Zhang, H., Chen, Y., Rong, W., Wang, J. and Tan, J. (2022). Effect of social media rumors on stock market volatility: A case of data mining in China. *Frontiers in Physics*, 10. doi:<https://doi.org/10.3389/fphy.2022.987799>.

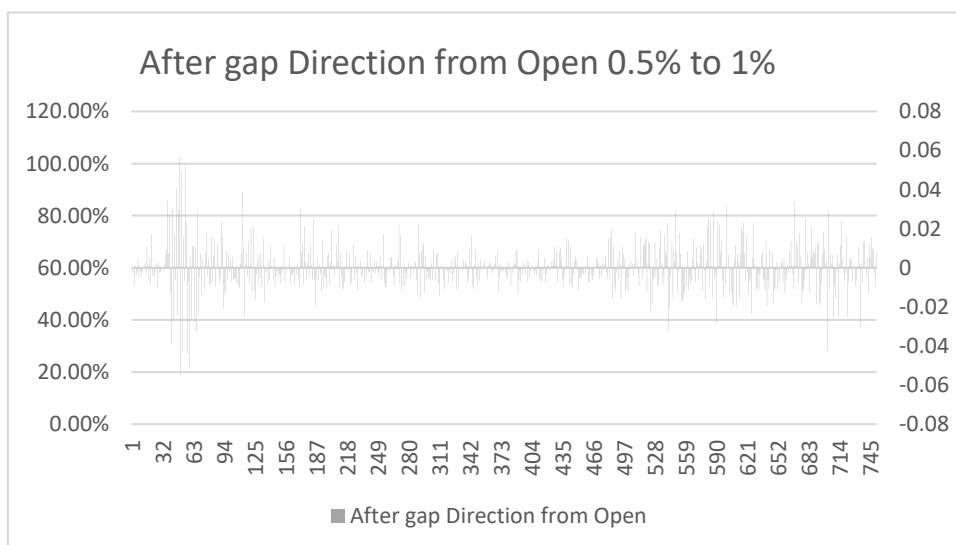
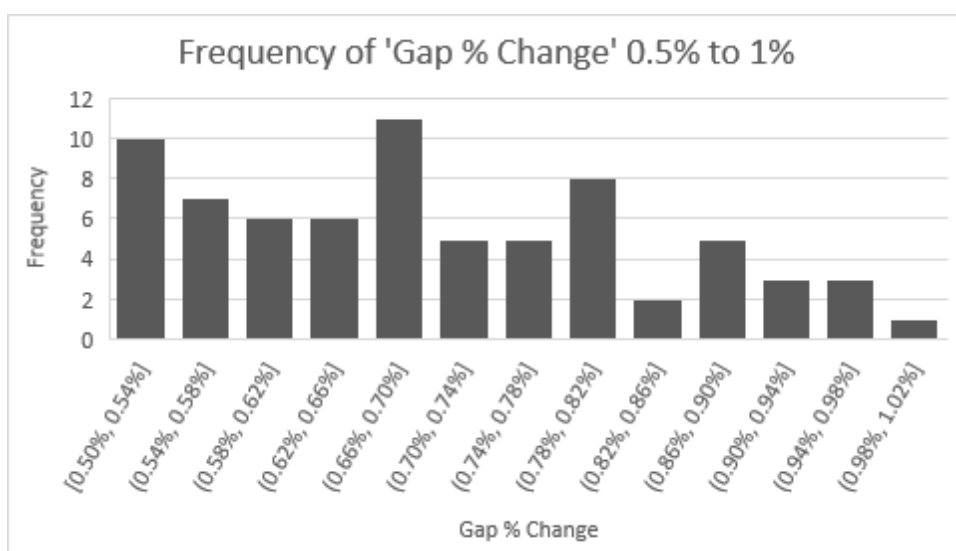
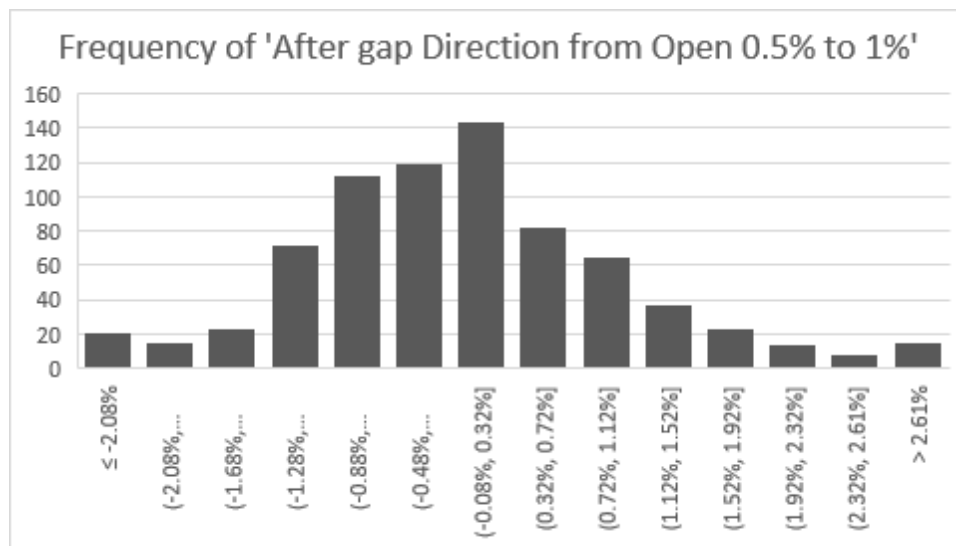
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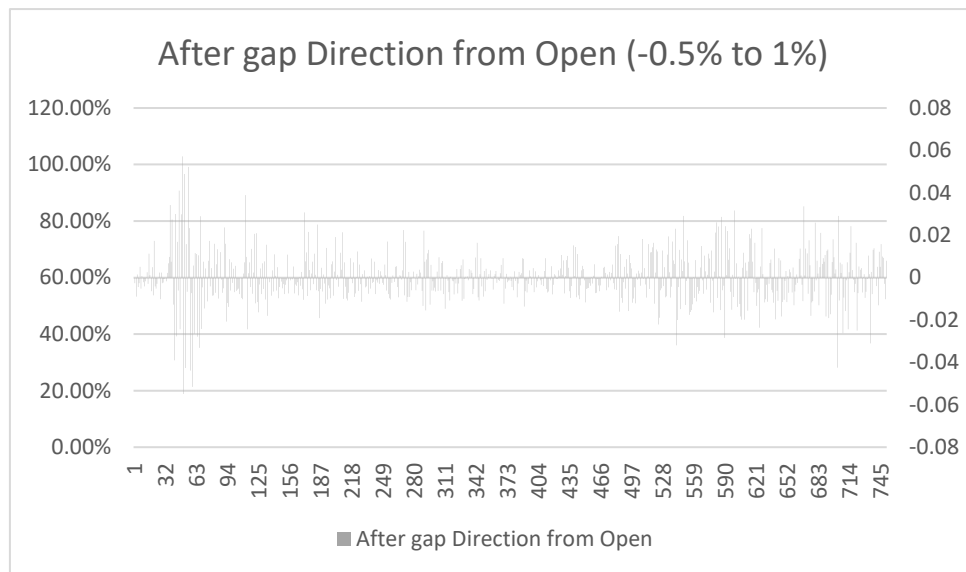
Gap > 2%		
Date	Gap % Change	After gap Direction from Open
3/10/2020	2.44%	-2.44%
3/13/2020	3.60%	-5.49%
3/24/2020	4.78%	-4.39%
4/6/2020	3.60%	-3.31%
4/7/2020	2.81%	2.89%
6/12/2020	2.30%	0.97%
6/16/2020	2.10%	0.20%
11/9/2020	2.10%	0.91%
11/10/2022	2.97%	-2.50%
Date	Gap % Change	After gap Direction from Open
Average	2.97%	-1.46%



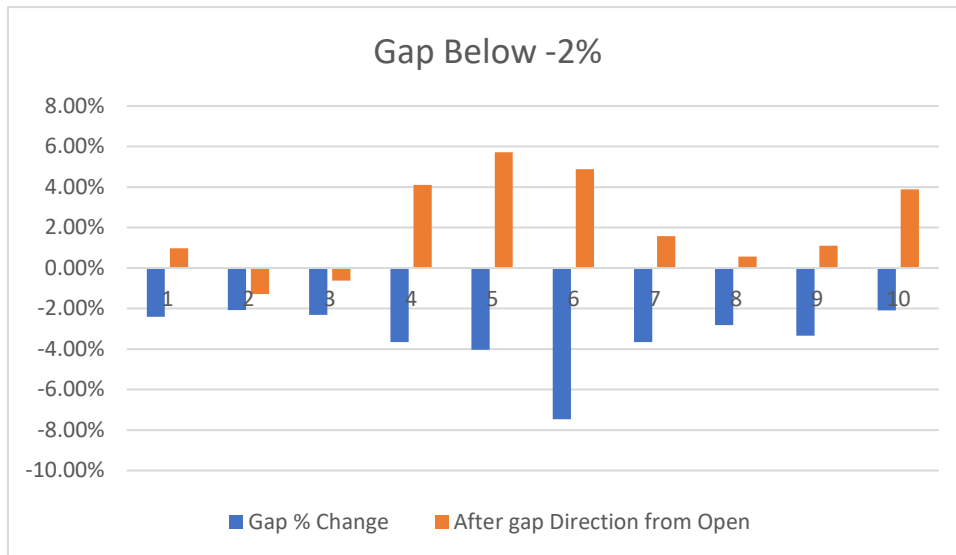
Gap Between 1% - 2%		
Date	Gap % Change	After gap Direction from Open
3/4/2020	1.41%	-2.77%
3/17/2020	1.66%	-4.27%
3/26/2020	1.04%	-5.15%
4/14/2020	1.57%	-1.46%
4/17/2020	1.53%	-1.13%
4/22/2020	1.88%	-0.41%
4/28/2020	1.09%	1.60%
4/29/2020	1.92%	-0.72%
5/7/2020	1.05%	-0.10%
5/18/2020	1.75%	-1.37%
5/20/2020	1.05%	-0.61%
5/26/2020	1.65%	0.41%
6/5/2020	1.65%	-0.95%
9/9/2020	1.14%	-0.86%
9/28/2020	1.07%	-0.53%
11/4/2020	1.11%	-1.09%
11/5/2020	1.23%	-0.71%
1/26/2022	1.19%	1.33%
3/9/2022	1.26%	-1.30%
5/10/2022	1.10%	0.85%
5/17/2022	1.10%	-0.91%
6/21/2022	1.10%	-1.33%
8/10/2022	1.42%	-0.70%
10/4/2022	1.31%	-1.73%
10/17/2022	1.55%	-1.08%
10/18/2022	1.86%	0.70%
11/4/2022	1.27%	-0.09%
11/15/2022	1.24%	0.37%
12/13/2022	1.98%	1.22%
Date	Gap % Change	After gap Direction from Open
Average	1.39%	-0.79%







Gap Below -2%		
Date	Gap % Change	After gap Direction from Open
2/24/2020	-2.40%	0.97%
2/28/2020	-2.08%	-1.28%
3/6/2020	-2.31%	-0.62%
3/9/2020	-3.65%	4.10%
3/12/2020	-4.03%	5.71%
3/16/2020	-7.47%	4.88%
3/18/2020	-3.66%	1.58%
3/27/2020	-2.82%	0.56%
4/1/2020	-3.35%	1.10%
6/11/2020	-2.09%	3.89%
Date	Gap % Change	After gap Direction from Open
Average	-3.39%	2.09%



Gap Category	Average Gap % Change	Average Post Gap Direction Change
Gaps > 2%	2.97%	-1.46%
Gaps 1% - 2%	1.39%	-0.79%
Gaps 0.5% - 1%	0.70%	-0.33%
Gaps -0.5% to 1%	-0.70%	0.18%
Gaps -1% to 2%	-1.37%	0.70%
Gaps < -2%	-3.39%	2.09%

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.991901913
R Square	0.983869406
Adjusted R Square	0.979836757
Standard Error	0.003179937
Observations	6

ANOVA					
	df	SS	MS	F	Significance F
Regression	1	0.002467085	0.00246709	243.975987	9.8103E-05
Residual	4	4.0448E-05	1.0112E-05		
Total	5	0.002507533			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	0.000494575	0.001300331	0.38034571	0.7230236	-0.00311572	0.00410487
Average Post Gap Di	-1.786526084	0.114376241	-15.6197307	9.8103E-05	-2.10408544	-1.46896673

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.060602779
R Square	0.003672697
Adjusted R Squ	0.000686701
Standard Error	0.010708596
Observations	1005

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	0.000423138	0.000141046	1.229973833	0.297551863
Residual	1001	0.114788711	0.000114674		
Total	1004	0.11521185			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.00242073	0.007965349	-0.30390738	0.761261618	-0.018051425	0.013209968	-0.0180514	0.013209968
Volume	6.09435E-13	3.18834E-13	1.91144994	0.056232113	-1.62244E-14	1.23509E-12	-1.622E-14	1.23509E-12
Gold	3.37552E-08	2.92636E-06	0.011534873	0.990799006	-5.70876E-06	5.77627E-06	-5.709E-06	5.77627E-06
USD	-7.1753E-06	5.68433E-05	-0.12622892	0.899576065	-0.000118721	0.000104371	-0.0001187	0.000104371